Formula Summary Sheet — Take-off Weight Sizing

Tables and figures numbered to match the *Source* column of the ASW example (Raymer, *Aircraft Design: A Conceptual Approach*)

Sizing equations

Take-off gross weight (Box 3.1)

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \dfrac{W_f}{W_0} - \dfrac{W_e}{W_0}} \tag{1}$$

Empty-weight fraction (Table 3.1)

$$\frac{W_e}{W_0} = A\, W_0^{\,C}\, K_{vs} \tag{2}$$

Composite aircraft

$$\left(\frac{W_e}{W_0}\right)_{comp} \approx 0.95 \left(\frac{W_e}{W_0}\right)_{metal} \tag{3}$$

Mission weight fraction (x segments)

$$\frac{W_x}{W_0} = \prod_{i=1}^{x} \frac{W_i}{W_{i-1}} \tag{4}$$

Fuel fraction (6% reserve + trapped fuel)

$$\frac{W_f}{W_0} = 1.06 \left(1 - \frac{W_x}{W_0}\right) \tag{5}$$

Cruise — Breguet range

$$\frac{W_i}{W_{i-1}} = \exp\left[\frac{-R\,C}{V\,(L/D)}\right] \tag{6}$$

Loiter — endurance

$$\frac{W_i}{W_{i-1}} = \exp\left[\frac{-E\,C}{(L/D)}\right] \tag{7}$$

Range / endurance (rearranged)

$$R = \frac{V}{C}\frac{L}{D}\ln\frac{W_{i-1}}{W_i}, \qquad E = \frac{1}{C}\frac{L}{D}\ln\frac{W_{i-1}}{W_i} \tag{8}$$

Propeller equivalent SFC (Table 3.4)

$$C = \frac{C_{bhp}\,V}{550\,\eta_p} \quad [1/\text{hr};\ V \text{ in ft/s}] \tag{9}$$

Wetted aspect ratio (Fig. 3.5 & 3.6)

$$A_{wet} = \frac{b^2}{S_{wet}} = \frac{A}{S_{wet}/S_{ref}} \tag{10}$$

Units: 1 nmi = 6076 ft = 1852 m; 1 ft = 0.3048 m; 1 lb = 0.4536 kg; SFC in 1/hr ÷3600 → 1/s; R, V, C, E must be in consistent units (exponents are dimensionless).

Table 3.1. Empty-weight fraction vs W_0, Eq. (2)

$W_e/W_0 = A\,W_0^C K_{vs}$	A	C
Sailplane — unpowered	0.86	−0.05
Sailplane — powered	0.91	−0.05
Homebuilt — metal/wood	1.19	−0.09
Homebuilt — composite	0.99	−0.09
General aviation — single engine	2.36	−0.18
General aviation — twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

$K_{vs} = 1.04$ variable sweep; $= 1.00$ fixed sweep. W_0 in lb.

Table 3.2. Historical segment weight fractions

Segment	W_i/W_{i-1}
Warm-up and take-off	0.970
Climb	0.985
Landing	0.995

Table 3.3. Specific fuel consumption, jet

Typical jet SFC (1/hr)	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

Table 3.5. L/D used in Eqs. (6)–(7)

	Cruise	Loiter
Jet	$0.866\,(L/D)_{max}$	$(L/D)_{max}$
Prop	$(L/D)_{max}$	$0.866\,(L/D)_{max}$

Table 3.4. Propeller specific fuel consumption, used with Eq. (9) (C_{bhp} in lb/hr/bhp)

Typical C_{bhp} / η_p	Cruise	Loiter
Piston-prop (fixed pitch)	0.4 / 0.8	0.5 / 0.7
Piston-prop (variable pitch)	0.4 / 0.8	0.5 / 0.8
Turboprop	0.5 / 0.8	0.6 / 0.8

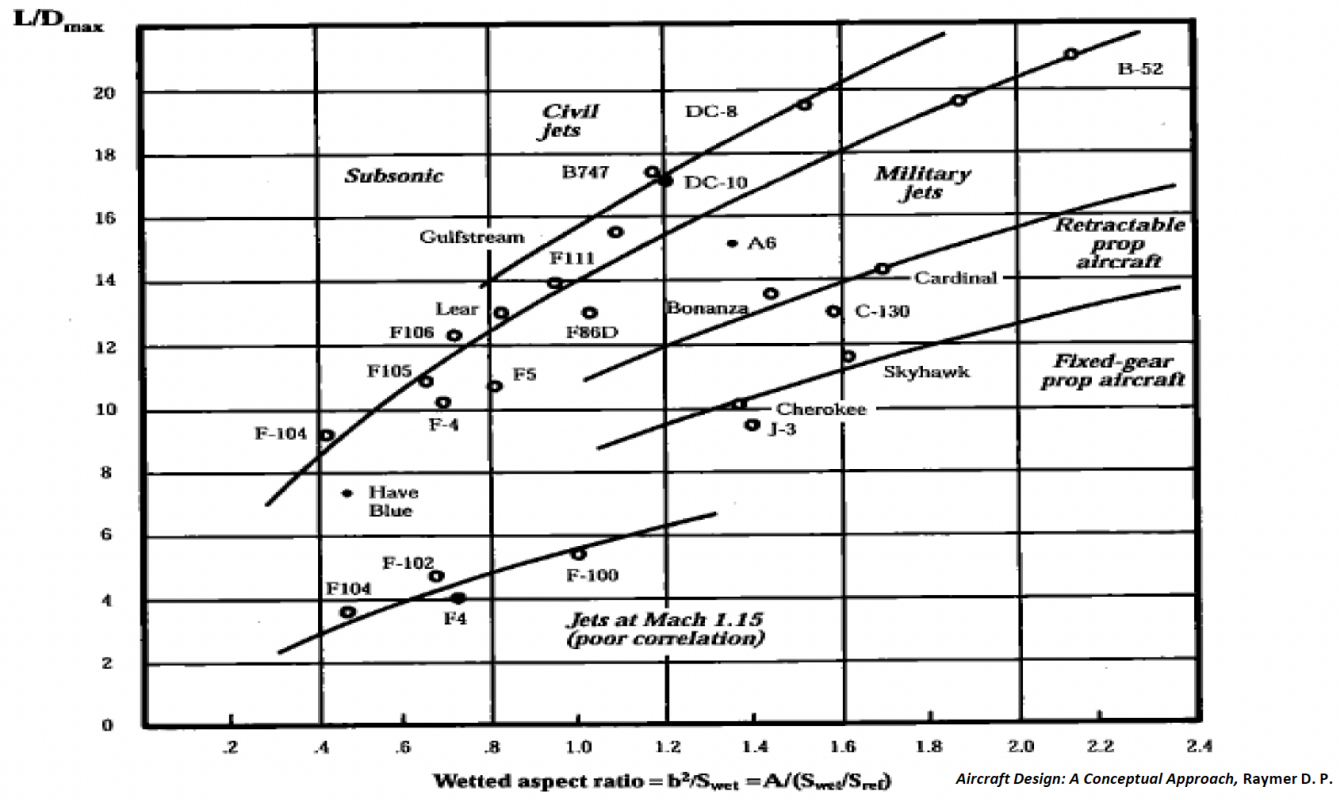


Figure 3.5. Maximum lift-to-drag ratio trends vs wetted aspect ratio, Eq. (10). (Raymer, *Aircraft Design: A Conceptual Approach*.)

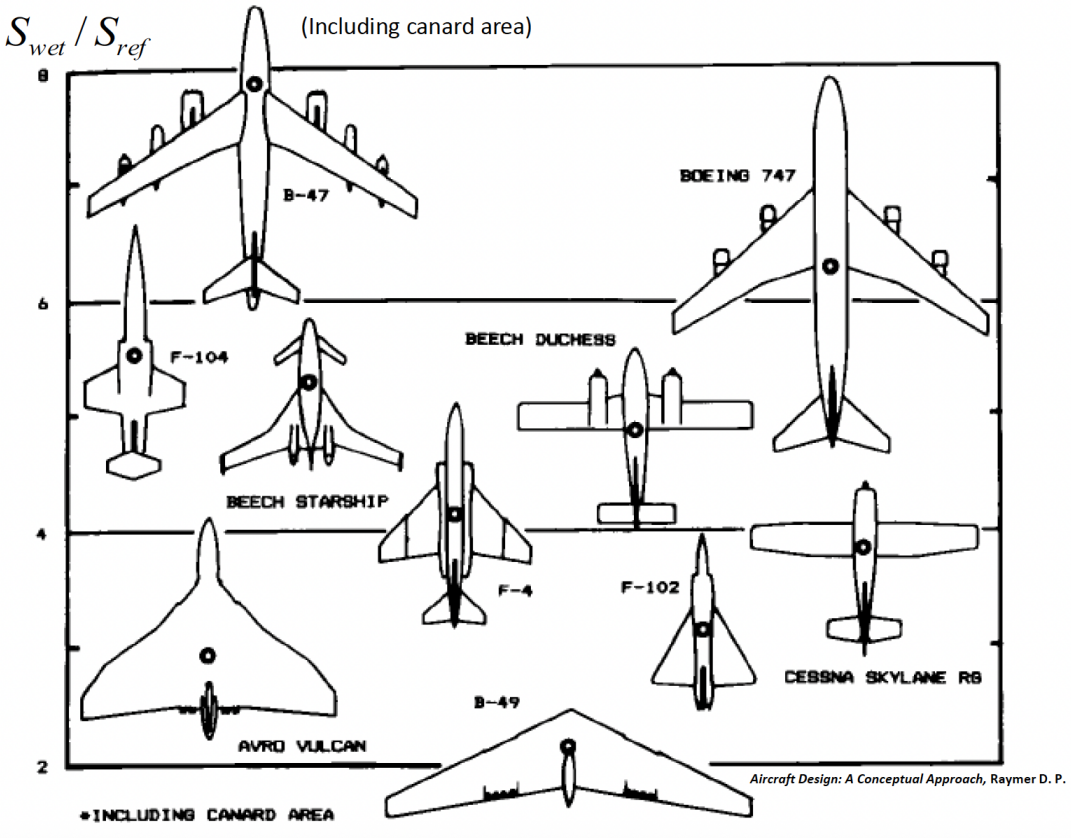


Figure 3.6. Wetted-area ratio S_{wet}/S_{ref} for representative configurations (including canard area). (Raymer, *Aircraft Design: A Conceptual Approach*.)